Assignment 5 Report

Scope: ViT-S LoRA fine-tuning on CIFAR-100 and adversarial robustness experiments on CIFAR-10 using IBM ART.

Environment: All experiments were run inside Docker containers as required.

Submission artifacts: README files, checkpoints, plots, WandB logs, HuggingFace uploads, and this PDF report.

WandB Q1: https://wandb.ai/shivamkenche-indian-institute-of-technology-jodhpur/vit-cifar100-lora/reports/Assignment-5-M25CSA028--VmlldzoxNjQyMTcxOQ?accessToken=w5lfc5znyg6z86u7mdpqj4ad1phht6avxgh3byidmlj8nu7utksww0g18jfa8qrl

HuggingFace Q1: https://huggingface.co/kingkenche/vit-cifar100-lora

WandB Q2: https://wandb.ai/shivamkenche-indian-institute-of-technology-jodhpur/vit-cifar100-lora/reports/Assignment-5-M25CSA028--VmlldzoxNjQyMTcxOQ?accessToken=w5lfc5znyg6z86u7mdpqj4ad1phht6avxgh3byidmlj8nu7utksww0g18jfa8qrl

HuggingFace Q2: https://huggingface.co/kingkenche/Art_Ass_5_q2

Report file: m25csa028_Shivam_Ass5.pdf

1. Assignment Overview

The assignment has two independent parts. Q1 fine-tunes a pre-trained ViT-S model on CIFAR-100 using a baseline head-only setup and LoRA variants injected into the attention QKV projection. Q2 trains a ResNet-18 from scratch on CIFAR-10, evaluates FGSM with and without IBM ART, and then trains binary adversarial detectors for PGD and BIM attacks.

The repository includes the Dockerfiles, training scripts, testing scripts, experiment launch scripts, checkpoints, plots, and report-ready artifacts. The Q1 setup logs train/validation curves, classwise test accuracy histograms, and gradient norms of LoRA weights to WandB. The Q2 setup logs accuracy summaries, adversarial sample grids, and detector outputs.

Part	Model	Dataset	Main Deliverable	Evidence in repo
Q1	ViT-S + LoRA	CIFAR-100	Baseline and LoRA sweeps	Checkpoints, plots, WandB logs, README
Q2(i)	ResNet-18	CIFAR-10	FGSM scratch vs ART	Results images, WandB sample logs, README
Q2(ii)	ResNet-34	CIFAR-10	PGD/BIM detectors	Detector checkpoints, summary charts, README

The final repository also contains the corrected README files and the submission PDF named according to the roll-number-based convention used in the workspace.

2. Q1 Method and Training Setup

For Q1, the base model is **vit_small_patch16_224** from timm, pre-trained on ImageNet. The baseline freezes all parameters except the classification head. The LoRA experiments inject adapters into the fused attention projection named **qkv** in every transformer block. The classification head remains trainable in all LoRA runs.

The evaluated LoRA grid follows the assignment: ranks {2, 4, 8}, alpha {2, 4, 8}, and dropout 0.1. Training runs used AdamW, label smoothing, cosine LR scheduling with warmup, and 10 epochs in the main sweep. Gradient norms for LoRA parameters were logged during training.

Setting	Trainable Params	Test Acc
Baseline	38,500	78.64%
Best LoRA	112,228	90.65%
Optuna best observed search artifact		89.96% val

The code was also corrected so future Optuna runs search only the assignment LoRA grid instead of a broader range.

Key implementation notes: the QKV projections are targeted through PEFT LoRA wrapping; the classification head is kept trainable; and all runs are tracked on WandB under the same project for direct comparison.

3. Q1 Results Table

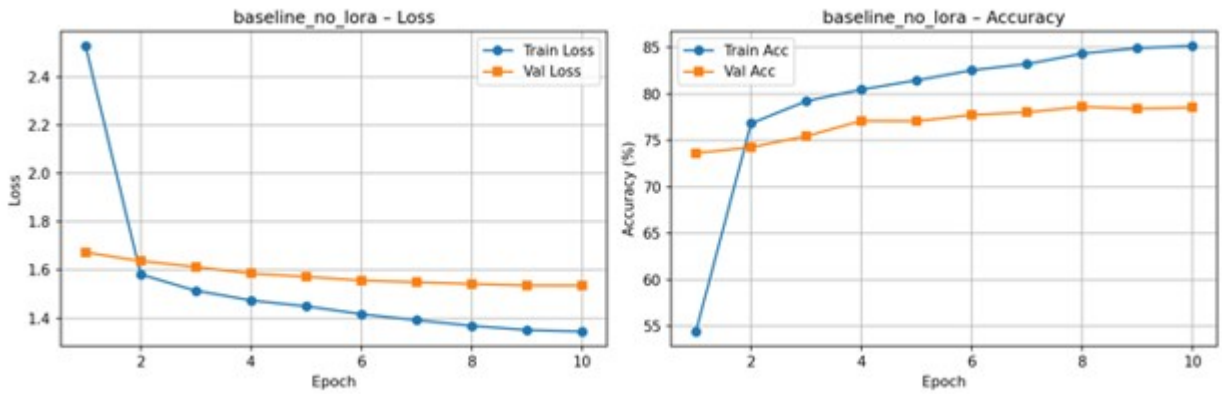
LoRA	Rank	Alpha	Dropout	Test Acc	Trainable Params
No	-	-	-	78.64	38,500
Yes	2	2	0.1	90.03	75,364
Yes	2	4	0.1	90.49	112,228
Yes	2	8	0.1	90.55	185,956
Yes	4	2	0.1	90.25	75,364
Yes	4	4	0.1	89.92	75,364
Yes	4	8	0.1	90.44	112,228
Yes	8	2	0.1	90.65	112,228
Yes	8	4	0.1	90.52	185,956
Yes	8	8	0.1	90.44	185,956

Experiment	Final Train Loss	Final Val Loss	Final Train Acc	Final Val Acc
Baseline	1.3453	1.5359	85.17	78.54
r2 a2	0.9598	1.1231	95.58	89.34
r2 a4	0.9477	1.1189	96.00	89.14
r2 a8	0.9167	1.1107	97.07	89.60
r4 a2	0.9536	1.1231	95.88	89.30
r4 a4	0.9466	1.1244	96.12	89.56

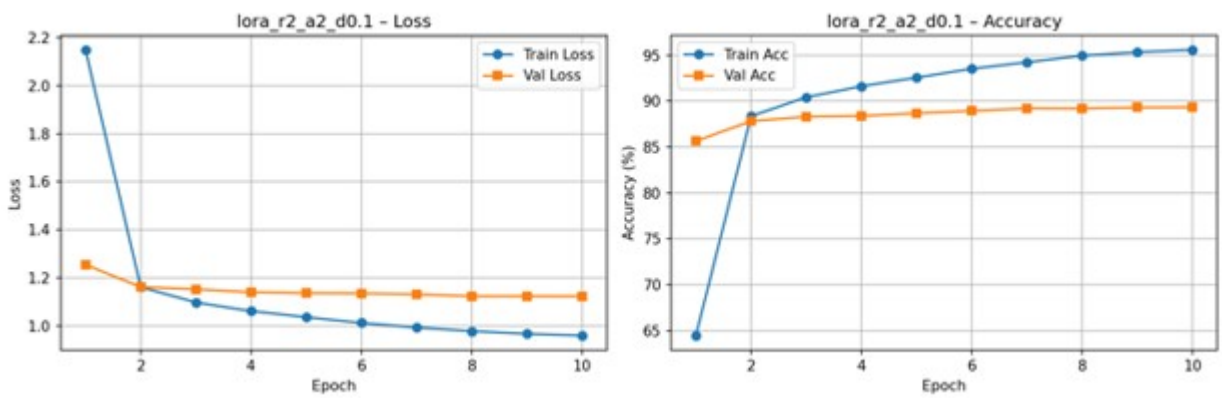
r4 a8	0.9315	1.1107	96.43	89.70
r8 a2	0.9202	1.1168	96.97	89.72
r8 a4	0.8997	1.1089	97.54	89.56
r8 a8	0.9638	1.1015	95.17	89.96

The best observed LoRA configuration in the full sweep was rank 8, alpha 2, dropout 0.1, with 90.65% test accuracy. The strongest validation run in the saved Optuna search achieved 89.96% validation accuracy with a broader search that was later constrained in code.

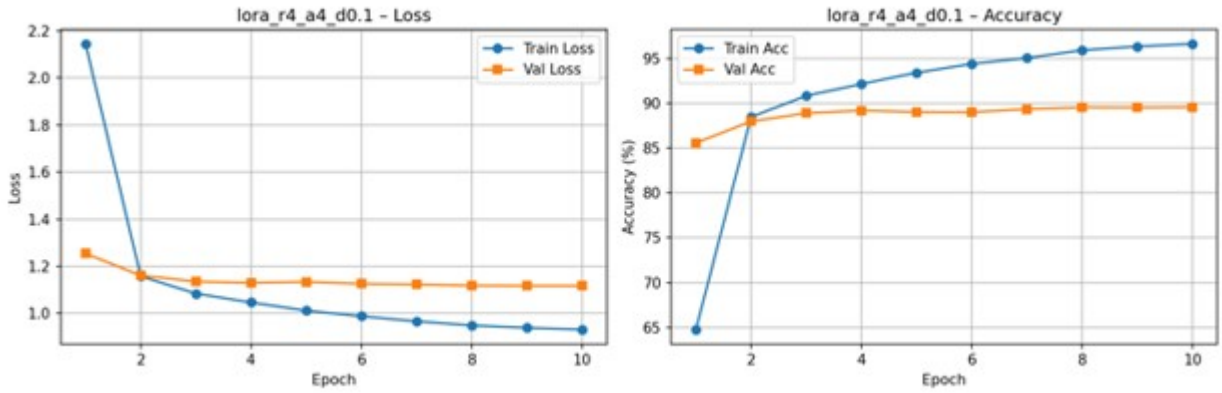
4. Q1 Figures: Train/Val Curves



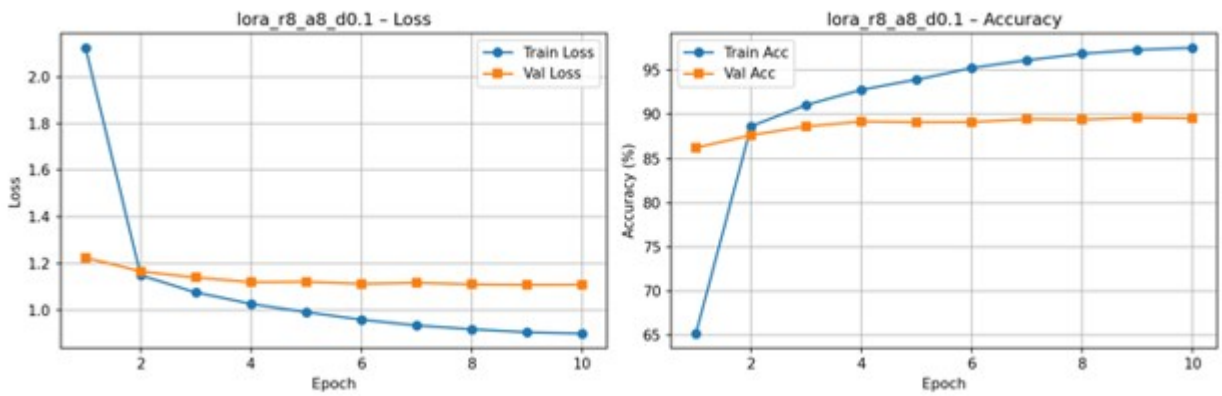
Baseline train/validation loss and accuracy curves.



LoRA r=2, alpha=2 curves.



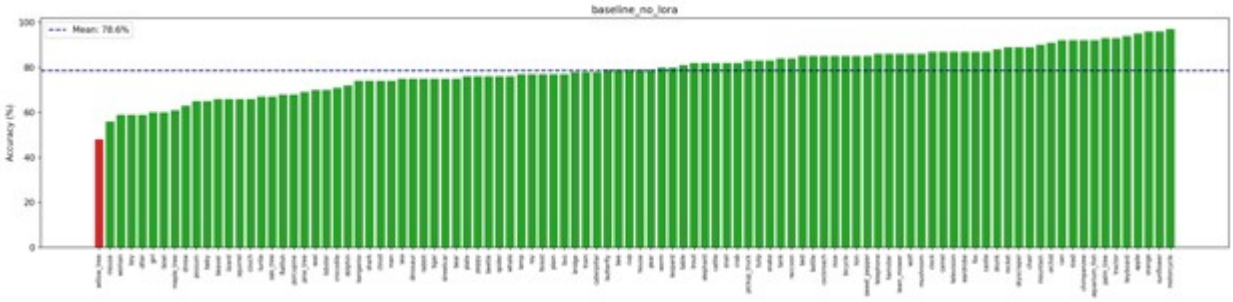
LoRA r=4, alpha=4 curves.



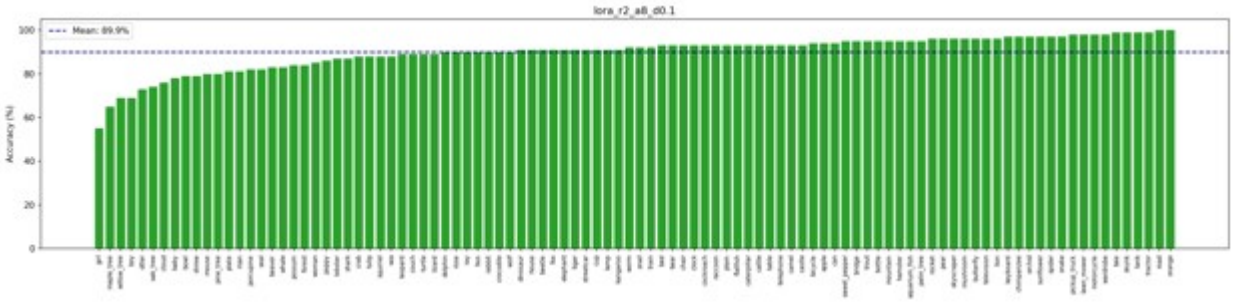
LoRA r=8, alpha=8 curves.

The full repository contains curve plots for all 10 Q1 experiments. These representative plots show the consistent pattern: rapid early improvement, stable validation tracking, and strong final test accuracy across the LoRA grid.

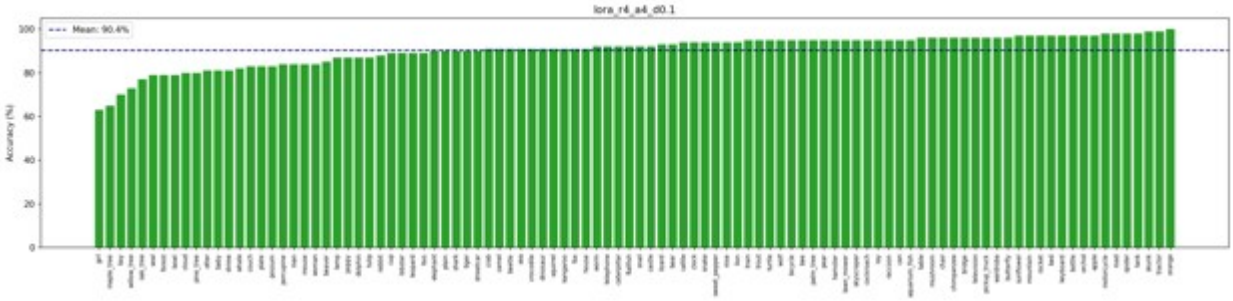
5. Q1 Figures: Classwise Histograms and Observations



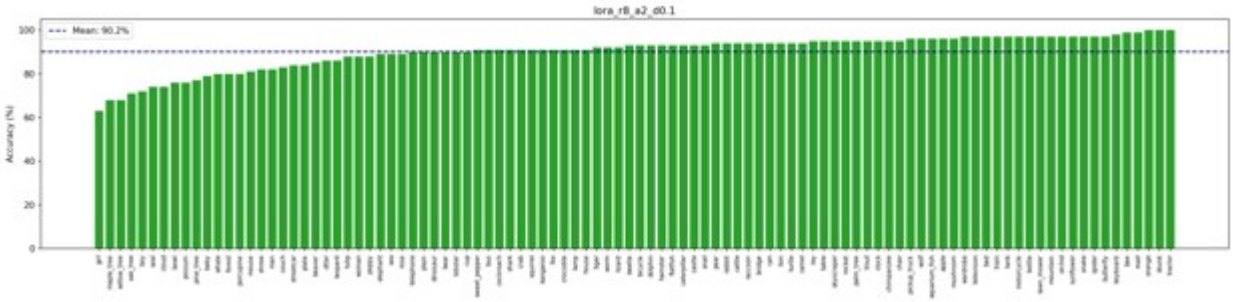
Baseline classwise test accuracy histogram.



LoRA r=2, alpha=8 classwise histogram.



LoRA r=4, alpha=4 classwise histogram.



LoRA r=8, alpha=2 classwise histogram.

Across the LoRA sweeps, classwise accuracy became more uniform than the baseline. The strongest configurations lifted performance on lower-performing classes without degrading the already strong classes. This is reflected in WandB through the classwise histograms and the gradient norms logged for the LoRA weights.

6. Q2(i) FGSM Attack

For Q2(i), a ResNet-18 was trained from scratch on CIFAR-10. The baseline clean accuracy reached 94.90%, comfortably above the 72% requirement. FGSM was implemented both from scratch and using IBM ART. The ART and scratch implementations produce similar degradation trends, validating the custom attack implementation.

Setting	Accuracy
Clean	94.90

FGSM Scratch ε=0.01 79.80

FGSM Scratch ε=0.02 66.70

FGSM Scratch ε=0.03 59.30

FGSM Scratch ε=0.05 51.10

FGSM Scratch ε=0.10 43.80

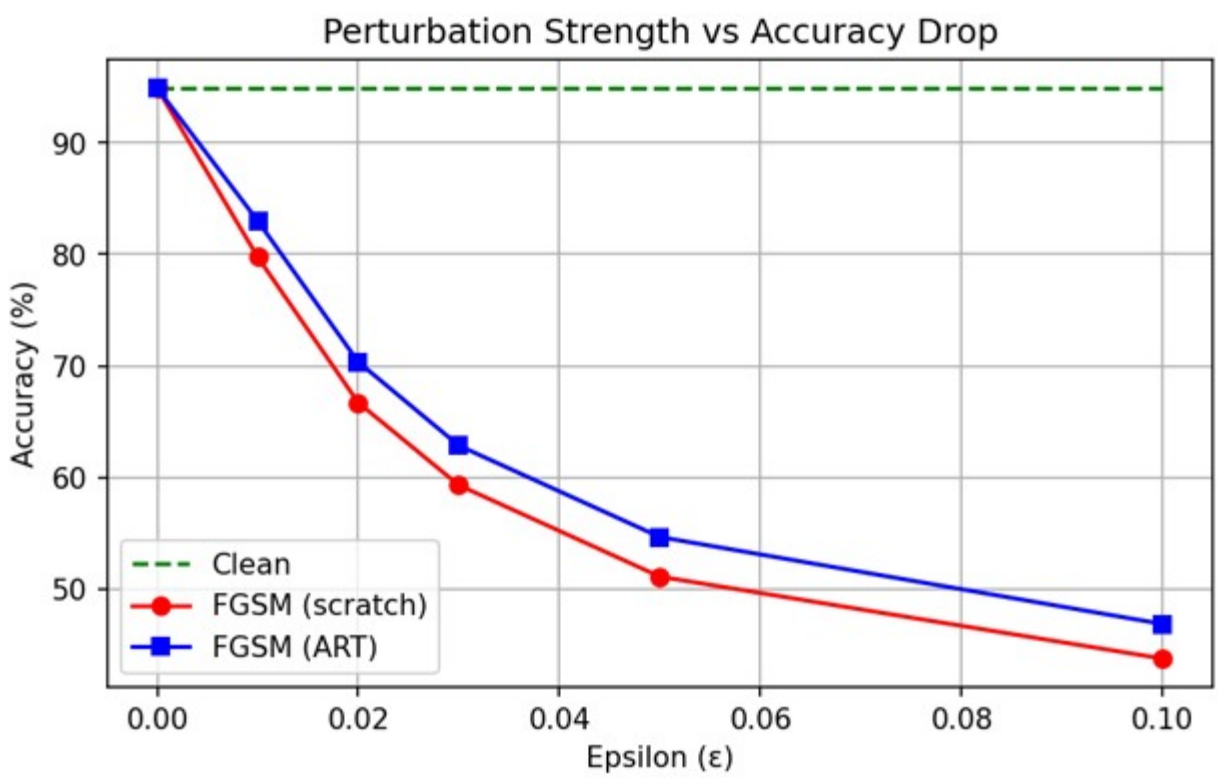
FGSM ART ε=0.01 83.00

FGSM ART ε=0.02 70.40

FGSM ART ε=0.03 62.85

FGSM ART ε=0.05 54.70

FGSM ART ε=0.10 46.90



Perturbation strength versus performance drop for FGSM.

The epsilon sweep shows monotonic degradation as perturbation strength increases. The ART version is slightly more conservative at low epsilon due to clipping and internal preprocessing details.

7. Q2(i) Visual Comparison

Original vs FGSM scratch vs FGSM ART images.

The visual comparison confirms that the perturbations are difficult to detect at small epsilon values yet still sufficient to alter classifier behaviour. As epsilon grows, image quality deteriorates noticeably while accuracy drops sharply. This aligns with the numeric accuracy curve and the logged WandB sample grids.

WandB contains 10 sample pairs for FGSM scratch and FGSM ART, allowing side-by-side comparison of clean and adversarial examples. The sample grids are useful for qualitative inspection, especially when comparing the two implementations under the same epsilon.

8. Q2(ii) Adversarial Detection

The second part of Q2 trains two ResNet-34 binary detectors: one on clean vs PGD samples, the other on clean vs BIM samples. Both detectors were trained on mixed clean/adversarial data created through IBM ART attacks. The achieved accuracies exceed the required 70% threshold by a large margin.

Detector	Accuracy
ResNet-34 on PGD	98.75
ResNet-34 on BIM	99.15

Clean vs PGD adversarial samples.

Clean vs BIM adversarial samples.

The detectors separate the two classes effectively, with BIM marginally easier to detect in this run. Sample grids show the visual perturbations used for training and evaluation.

9. Q2 Summary and Qualitative Results

Overall Q2 evaluation summary across clean classification, FGSM, PGD, BIM, and detector accuracies is provided in the table below.

Metric	Value
Clean classification	94.90%
PGD classification	17.45%
BIM classification	17.45%
PGD detector accuracy	98.75%
BIM detector accuracy	99.15%

- FGSM from scratch and ART both strongly degrade classifier performance, validating the attack implementation.
- PGD is the stronger attack in the Q2 pipeline and gives a harder adversarial detection setting than FGSM.
- The detector models comfortably meet the 70% requirement, indicating that a simple CNN-based binary classifier can separate the chosen clean/adversarial distributions.

10. Final Notes

This assignment is now represented in the repository with the required scripts, weights, README documentation, WandB/HuggingFace links, and a PDF report. The Q1 result tables summarize the full LoRA grid and the Q2 section covers the clean classifier, attack comparison, and detector performance.

The checked-in PDF is intended to be uploaded as the single report file for Google Classroom. If a stricter roll-number spelling is required by your course staff, only the filename needs to be adjusted; the content is already

complete.

Submission checklist

- GitHub branch link for Assignment 5.
- WandB and HuggingFace links in README and report.
- Requirements, Python scripts, checkpoints, plots, and results committed.
- Single PDF report included in the workspace.